

Technology #14

Acoustic Obstruction Detection for CSF Shunts

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Acoustic Obstruction Detection for CSF Shunts

WHY DOES IT MATTER?

Shunt obstruction is detected late (symptoms then imaging)

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 11 design inputs, 8 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Acoustics + Ultrasound + Signal Detection. 7-step build plan with specific test articles and acceptance criteria. Target buyer: Neuroshunt manufacturer; ultrasound device company.

WHAT DOES THE BUYER GET?

Acoustic obstruction detection concept + mechanism description; Acoustic impedance + reflection + TOA governing equations; Bench test protocols (detection sensitivity, false positives, acoustic output, EMC); External precedent catalog (IVUS, Doppler, AIUM/NEMA UD-2); Critical UNKNOWN disclosure (tissue obstruction sensitivity, optimal frequency, false positive rate); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production transducer design (frequency, geometry, material); Production drive + receive electronics + signal processor; Production classifier (validated in vivo); Manufacturing process for catheter-integrated acoustic detector; Regulatory submission (510(k) with IVUS predicate, or De Novo); Clinical validation (sensitivity + specificity vs imaging)

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Acoustic transducer prototypes (various frequencies + geometries)

WHAT WOULD MAKE US KILL IT?

Tissue obstruction acoustic impedance is too similar to CSF for reliable detection (Z ratio < 1.1)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Acoustic transducer prototypes (various frequencies + geometries)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Acoustic transducer prototypes (various frequencies + geometries) - 12 weeks

Basis: Build plan WP. Scope: Sensitivity, bandwidth, directivity. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Classifier algorithm + training data - 10 weeks

Basis: Build plan WP. Scope: Classifier accuracy (sensitivity + specificity). Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens (PZT transducer + parylene encapsulation + acoustic window) - 12 weeks (external)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: CSF is approximately water-like acoustically ($Z \sim 1.5 \text{ MRayl}$); Obstruction has different acoustic impedance (debris $\sim$ tissue-like, air $\sim$ vacuum-like); Catheter lumen is acoustically transparent at chosen frequency; Multipath and reverberation manageable; CSF attenuation comparable to water at operating frequency. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Acoustic transducer prototypes (various frequencies + geometries). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Catheter-integrated acoustic transducer detects obstruction via pulse-echo impedance contrast

POTENTIAL VALUE DRIVER: Addresses: Shunt obstruction is detected late (symptoms then imaging)

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Neuroshunt manufacturer; ultrasound device company

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Acoustics + Ultrasound + Signal Detection

Disciplines: Acoustic engineering (transducer design), Ultrasound physics (wave propagation, impedance), Signal processing (classification, detection), Catheter integration (miniaturization)

Architecture: Catheter-integrated acoustic transducer that emits ultrasonic pulse and detects reflections from obstruction (debris, tissue, air bubble). Signal classifier identifies obstruction type and severity.

Subsystems:

SS-01: Acoustic transducer (piezoelectric) - Emits ultrasonic pulse and receives reflected echoes

SS-02: Drive + receive electronics - Generates transmit pulse; amplifies and digitizes received echo

SS-03: Signal classifier (ML or threshold-based) - Classifies echo pattern as normal, obstructed, or specific obstruction type

SS-04: Catheter integration (acoustic window) - Acoustic coupling between transducer and CSF/lumen

2. Mechanism Architecture

Physical Changes: Acoustic pulse propagates through CSF in catheter lumen; reflections from obstructions return to transducer; classifier identifies obstruction presence and type

Key Physics: Acoustic impedance mismatch at obstruction boundary causes reflection. Reflection coefficient $R = (Z_2 - Z_1) / (Z_2 + Z_1)$. Amplitude of reflection determines detection; time-of-flight determines distance.

3. Governing Engineering Model

Summary: Pulse-echo ultrasound detection. Acoustic impedance contrast at obstruction boundary causes reflection; transducer receives echo; classifier identifies obstruction.

Equations:

$Z = \rho \cdot c$ [acoustic impedance of CSF in catheter lumen – P-28 obstruction detection]

$R = (Z_2 - Z_1) / (Z_2 + Z_1)$ [reflection at CSF-obstruction interface in P-28 catheter]

$f \cdot \lambda = c$ [wave equation]

$TOF = 2 \cdot d / c$ [round-trip time of flight]

Attenuation: $A(d) = A_0 \cdot \exp(-\alpha \cdot f \cdot d)$ [frequency-dependent attenuation]

Assumptions:

- CSF is approximately water-like acoustically ($Z \sim 1.5 \text{ MRayl}$)
- Obstruction has different acoustic impedance (debris ~ tissue-like, air ~ vacuum-like)
- Catheter lumen is acoustically transparent at chosen frequency
- Multipath and reverberation manageable
- CSF attenuation comparable to water at operating frequency

Boundary Conditions:

- Frequency: 1-50 MHz (low for deep penetration, high for IVUS-like resolution)
- Catheter ID: ~1-2 mm
- Detection range: 0-50 mm (length of catheter segment)
- Acoustic coupling: CSF fills lumen

Failure Regimes:

- Low impedance contrast (tissue obstruction) -> weak reflection -> poor SNR
- Attenuation excessive at high frequency -> no echo from distant obstruction

- Multipath/reverberation -> false positives
- Transducer failure -> no signal
- Classifier misclassification -> wrong obstruction type

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Obstruction causes 30-50% of shunt failures; early detection enables preemptive intervention (R332)	VERIFIED
DI-002	Functional requirement	Acoustic obstruction detection via catheter-integrated transducer + classifier	MODELLED
DI-003	Acoustic impedance CSF	EXTERNAL_PRECEDENT (~1.5 MRayl)	EXTERNAL_PRECEDENT
DI-004	Acoustic impedance tissue/debris	EXTERNAL_PRECEDENT (~1.6-1.7 MRayl soft tissue)	EXTERNAL_PRECEDENT
DI-005	Acoustic impedance air	EXTERNAL_PRECEDENT (~0.0004 MRayl)	EXTERNAL_PRECEDENT
DI-006	Catheter diameter	~2 mm (standard shunt catheter)	VERIFIED
DI-007	Detection range	0-50 mm (catheter segment length)	MODELLED
DI-008	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-009	Sterilization	UNKNOWN	UNKNOWN
DI-010	Acoustic output safety (AIUM/NEC/IEC)	UNKNOWN — must comply with safety limits	UNKNOWN
DI-011	EMC	UNKNOWN — required IEC 60601-1-2	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Acoustic transducer design (frequency, geometry, material)	ABSENT	Frequency; Transducer geometry; Material (PZT or PVDF)
DO-002	Drive + receive electronics	CONCEPTUAL	
DO-003	Classifier algorithm	MODELLED	
DO-004	Catheter integration (acoustic window)	ABSENT	Window geometry; Material; Coupling design

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Operating frequency	UNKNOWN — design choice (likely 1-20 MHz for B-mode scale resolution) trade-off	Hz	Resolution vs. penetration	Bench SNR test
Transducer size	UNKNOWN — constrained by catheter ~2 mm	mm	Catheter geometry	Geometric verification with catheter vendor
Acoustic impedance CSF	EXTERNAL_PRECEDENT (~1.5 MRayl), similar to water at 37°C	MRayl	Published acoustic properties	
Acoustic impedance obstruction	EXTERNAL_PRECEDENT (~1.6 MRayl soft tissue)	MRayl	Published tissue acoustic properties	
Acoustic impedance air	EXTERNAL_PRECEDENT (~0.0004 MRayl)	MRayl	Published acoustic properties	
CSF attenuation coefficient	EXTERNAL_PRECEDENT (~0.02 dB/cm/MHz) ² similar to water	dB/cm/MHz ²	Published water/CSF acoustic properties	
Classifier accuracy target	UNKNOWN — clinical utility threshold TBD	%	UNKNOWN	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Low impedance contrast (tissue obstruction)	Similar to tissue obstruction	Higher CSF transmit power (limited by safety with tissue Doppler)	Known achievable sensitivity for tissue obstruction	
Attenuation limits detection	High frequency pulses attenuated	Lower frequency (trade-off with penetration); multiple frequencies	Known optimal frequency	
Multipath / reverberation	Catheter walls and structures	Signal processing (e.g. gating), catheter design	Known multipath severity in vivo	
Transducer failure	Piezoelectric element degradation	Redundant transducer; self-test	Reliability test	UNKNOWN failure rate
Classifier misclassification	Minor threshold classifier produces wrong result	Conservative thresholds; prospective	Known achievable accuracy	
Acoustic window failure	CSF does not fill lumen (air bubble, dry)	Design ensures CSF coupling; air bubble detection	UNKNOWN coupling reliability in vivo	
False positives (clinical significance)	Classified as normal variant	Conservative threshold; confirmatory study	Known false positive rate clinically	
EMC interference	External EM affects electronics	Shielding + IEC 60601-1-2	EMC test	UNKNOWN specific interferences

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Low impedance contrast (tissue simulation)	CSF -> weak reflector	Higher power (limited); better classifier	UNKNOWN achievable sensitivity
Attenuation limits range	High freq attenuates before reaching obstacle	Lower frequency (tradeoff with resolution)	UNKNOWN optimal frequency
Multipath / reverberation	Catheter walls cause spurious reflections	Signal processing + acoustic windows	UNKNOWN in vivo severity
Transducer failure	Piezo element degrades or disconnects	Redundant transducer + self-test	UNKNOWN failure rate
Classifier misclassification	ML or threshold produces wrong result	Diverse training + conservative threshold	UNKNOWN achievable accuracy
Acoustic window failure	CSF doesn't fill lumen (air bubble)	Design ensures coupling + air bubble detection	UNKNOWN coupling reliability
False positives	Classifier flags normal variation as obstruction	Conservative threshold + confirmatory imaging	UNKNOWN clinical FP rate
EMC interference	External EM affects electronics	Shielding + IEC 60601-1-2	UNKNOWN specific interferences

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Detection sensitivity $\geq 90\%$	Bench phantom test	$\geq 90\%$ detection at clinically relevant sizes	NOT TESTED
V-002	False positive $< 5\%$	Bench normal condition test	$< 5\%$	NOT_TESTED
V-003	Acoustic output safety	AIUM/NEMA UD-2	Within safety limits	NOT_TESTED
V-004	Biocompatibility ISO 10993	ISO 10993	Pass	NOT_TESTED
V-005	EMC IEC 60601-1-2	EMC test	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Clinical obstruction detection accuracy	Clinical study vs imaging	Sensitivity + specificity clinically useful	NOT PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Acoustic transducer	PZT (lead zirconate titanate)	Standard ultrasound transducer material	ISO 10993 (with encapsulation details)	CANDIDATE — acoustic characterization required
Acoustic transducer	PVDF (polyvinylidene fluoride)	Polymer piezoelectric; biocompatible	ISO 10993 + acoustic characterization	CANDIDATE — preferred for biocompatibility
Catheter body	Silicone elastomer	Standard CSF shunt material	ISO 10993 + acoustic transparency	CANDIDATE
Acoustic window	Polyurethane (acoustically transparent)	Standard ultrasound window material	ISO 10993 + acoustic characterization	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Acoustic transducer (piezoelectric)	1	CUSTOM_COMPONENT	PZT or PVDF	UNKNOWN — custom fabrication	CRITICAL	Acoustic characterization + biocompatibility
02	Drive + receive electronics	1	CUSTOM_COMPONENT	PCB + components	Multiple	HIGH	Circuit simulation + bench
03	Signal processor / MCU	1	COTS_CANDIDATE	MCU (COTS)	Multiple	HIGH	Software V&V per IEC 62304
04	Catheter body with acoustic window	1	CUSTOM_COMPONENT	Silicone elastomer + polyurethane	UNKNOWN — multiple options	HIGH	Acoustic transparency + biocompatibility

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no catheter-integrated acoustic obstruction detector process validated

Candidate Processes:

- Piezoelectric transducer microfabrication: Standard ultrasound transducer manufacturing - Custom for catheter-scale size
- Hybrid circuit assembly (transducer + electronics): Standard hybrid assembly - Miniaturization required
- Catheter extrusion + acoustic window integration: Standard catheter manufacturing - Custom for acoustic window
- Hermetic encapsulation (if PZT used): Standard implantable encapsulation - Tradeoff between encapsulation and acoustic performance

14. External Evidence

Source 1: Acoustically Active Catheter for Intracardiac Navigation by Color

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5515670>

Excerpt: Jun 5, 2017 · We present a catheter fitted with a piezoelectric crystal, which vibrates and produces an instantaneous marker in color flow Doppler scans.

Source 2: Acoustic Wave Catheter - Active Life Medical Products

URL: <https://activelifemed.com/glossary/acoustic-wave-catheter-learn-more-in-our-activelifemed-glossary>

Excerpt: Oct 16, 2024 · An acoustic wave catheter is a specialized medical device designed to deliver therapeutic ultrasound energy to targeted tissues within the body.

Source 3: Mechanisms of Focused Ultrasound

URL: <https://www.fusfoundation.org/the-technology/mechanisms-of-focused-ultrasound>

Excerpt: Neuromodulation with focused ultrasound allows noninvasive stimulation or suppression of neural activity. Low-intensity ultrasound can reversibly alter brain or ...

Source 4: A flexible catheter-based sensor array for upper airway soft tissues

URL: <https://www.nature.com/articles/s41467-024-55088-y>

Excerpt: Jan 2, 2025 · Here, we report a catheter-based flexible pressure sensor array that continuously monitors soft measuring upper airway obstruction location^{1,2}. ...

Source 5: Advanced Urology Therapy Device In Tyler, TX | UroShield

URL: <https://nanovibronix.com/uroshield-product-information>

Excerpt: The ultrasonic waves generated by the clip create an acoustic shield on the surfaces of the catheter to interfere with the attachment of bacteria, prevent ...

15. Transfer Boundary

YOU RECEIVE:

- + Acoustic obstruction detection concept + mechanism description
- + Acoustic impedance + reflection + TOA governing equations
- + Bench test protocols (detection sensitivity, false positives, acoustic output, EMC)
- + External precedent catalog (IVUS, Doppler, AIUM/NEMA UD-2)
- + Critical UNKNOWN disclosure (tissue obstruction sensitivity, optimal frequency, false positive rate)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production transducer design (frequency, geometry, material)
- > Production drive + receive electronics + signal processor
- > Production classifier (validated in vivo)
- > Manufacturing process for catheter-integrated acoustic detector
- > Regulatory submission (510(k) with IVUS predicate, or De Novo)
- > Clinical validation (sensitivity + specificity vs imaging)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable detection sensitivity for tissue obstructions (low impedance contrast) — UNKNOWN (critical)
- Optimal frequency balancing resolution and attenuation — UNKNOWN
- False positive rate in clinical use — UNKNOWN
- Acoustic coupling reliability in vivo — UNKNOWN
- Classifier accuracy in vivo — UNKNOWN
- Regulatory pathway (likely 510(k) with IVUS predicate, or De Novo) — UNKNOWN
- Clinical utility threshold for obstruction size — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Acoustic transducer prototype (options, full type)	Acoustic phantom, full type	Sensitivity, bandwidth, directivity	Sensitivity + bandwidth + directivity	Final test report	12 weeks
WP-02	Acoustic phantom with obstruction	Phantom transducer, Data acquisition	Detection sensitivity for ~90% obstruction	Detection sensitivity	Final test report	8 weeks
WP-03	Classifier algorithm + training data	Training datasets from WP-02	Classification accuracy (sensitivity + specificity)	Sensitivity + specificity	Classifier report	10 weeks
WP-04	Acoustic output measurement	AIUM/NEMA UD-2 mechanical system	Acoustic output within safety limits	Within safety limits	Acoustic output report	8 weeks
WP-05	Biocompatibility specification	ISO 10993 lab	ISO 10993 series capsule	Pass + acoustic window	ISO 10993 report	12 weeks (external)
WP-06	EMC test articles (acoustic transducer + drive electronics)	EMC lab	EMC emissions	Pass	EMC report	8 weeks (external)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-07	In vivo obstruction detection	Animal facility	Detection accuracy	clinically useful sensitivity	in vivo specificity	20 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: Neuroshunt manufacturer; ultrasound device company.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Tissue obstruction acoustic impedance is too similar to CSF for reliable detection (Z ratio < 1.1)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Tissue obstruction acoustic impedance is too similar to CSF for reliable detection (Z ratio < 1.1)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.